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**Supplementary table 1A. Baseline characteristics of the oral and intravenous glucocorticoids.**

| Parameters | Oral steroid (n=92) | Intravenous steroid (n=61) |
|---|---|---|
| **Demographic characteristics** | | |
| Age (years) | 41 (36.5-47.5) | 40 (28-48) |
| Female gender | 47/71 (66.2) | 42/56 (75) |
| Duration of symptoms (months) | 8 (3-8.8) | 4 (1-14) |
| **Clinical features** | | |
| Headache | 58/90 (64.4) | 37/55 (67.3) |
| Visual deficits | 23/90 (25.5) | 14/55 (25.45) |
| Cranial nerve palsy | 10/90 (11.1) | 8/55 (14.54) |
| **Anterior pituitary dysfunction** | | |
| Hypocortisolism | 42/70 (60) | 38/61 (62.3) |
| Hypothyroidism | 45/70 (64.3) | 30/31 (49.2) |
| Hypogonadotropism | 47/69 (68.1) | 32/59 (54.2) |
| Hyposomatotropism | 19/65 (29.2) | 23/47 (48.9)* |
| Hypoprolactinemia | 8/63 (12.7) | 5/53 (9.4) |
| Hyperprolactinemia | 23/62 (37.1) | 24/53 (45.3) |
| **Diabetes Insipidus** | 36/71 (50.7) | 25/60 (41.7) |
| **Radiological features** | | |
| Pituitary enlargement | 57/70 (81.4) | 51/60 (85) |
| Thickened stalk | 51/57 (89.5) | 45/49 (91.8) |

Data are expressed as % (n/N) or median (interquartile range)

* p value <0.05 between oral steroid and intravenous steroid groups.

**Supplementary table 1B. Baseline characteristics medium, high, and very high-dose glucocorticoids.**

| Parameters | Medium dose (n=16) | High dose (n=44) | Very high dose (n=53) |
|---|---|---|---|
| **Baseline characteristics** | | | |
| Age (years) | 38 (32.5-42) | 40.5 (34.5-54.5) | 41 (28-48) |
| Female gender | 10/16 (62.5) | 29/44 (65.9) | 35/48 (72.9) |
| Duration of symptoms (months) | 5 (2-8) | 4 (2-12) | 3 (1-14) |
| **Clinical features** | | | |
| Headache | 12/16 (75) | 28/44 (63.6) | 32/47 (68.1) |
| Visual deficits | 5/15 (33.3) | 12/44 (27.3) | 12/47 (25.5) |
| Cranial nerve palsy | 3/16 (18.8)* | 1/44 (2.3) | 7/47 (14.9)# |
| **Anterior pituitary dysfunction** | | | |
| Hypocortisolism | 10/16 (62.5) | 27/44 (61.4) | 33/53 (62.3) |
| Hypothyroidism | 11/16 (68.8) | 29/44 (65.9) | 25/53 (47.2) |
| Hypogonadotropism | 11/16 (68.8) | 28/43 (65.1) | 29/51 (56.9) |
| Hyposomatotropism | 5/15 (33.3) | 13/40 (32.5) | 20/39 (51.3) |
| Hypoprolactinemia | 4/16 (25) | 5/37 (13.5) | 4/46 (8.7) |
| Hyperprolactinemia | 12/16 (75) | 22/35 (62.9) | 24/46 (52.2) |
| **Diabetes Insipidus** | 8/16 (50) | 24/44 (54.5) | 21/53 (39.6) |
| **Radiological features** | | | |
| Pituitary enlargement | 14/16 (87.5) | 32/44 (72.7) | 45/53 (84.9) |
| Thickened stalk | 12/12 (100) | 32/38 (84.2) | 41/43 (95.3) |

Data are expressed as % (n/N) or median (interquartile range).
*p value of <0.05 between medium and high dose glucocorticoids.
#p value of <0.05 between very high dose and high dose glucocorticoids.

**Table 1C. Baseline characteristics of varied glucocorticoid treatment duration.**

| **Parameters** | **<6.5 weeks (n=28)** | **6.5-12 weeks (n=58)** | **>12 weeks (n=25)** |
|---|---|---|---|
| **Baseline characteristics** | | | |
| Age (years) | 41 (31.5/54) | 41 (38-41) | 39 (35-51) |
| Female gender | 20/28 (71.4) | 29/37 (78.4) | 13/25 (52) |
| Duration of symptoms (months) | 4 (1.5-12) | 8.8 (2-8.8) | 6 (3.5-12) |
| **Clinical features** | | | |
| Headache | 17/28 (60.7) | 41/55 (74.5) | 15/25 (60) |
| Visual deficits | 8/28 (28.6) | 12/56 (21.4) | 3/24 (12.5) |
| Cranial nerve palsy | 2/28 (7.1) | 9/55 (16.4) | 2/25 (8) |
| **Anterior pituitary dysfunction** | | | |
| Hypocortisolism | 18/28 (64.3) | 24/37 (64.9) | 13/25 (52) |
| Hypothyroidism | 12/28 (42.9) | 21/37 (56.8) | 15/25 (60) |
| Hypogonadotropism | 17/28 (60.7) | 22/35 (62.9) | 19/25 (76) |
| Hyposomatotropism | 9/25 (36) | 11/27 (40.7) | 9/22 (40.9) |
| Hypoprolactinemia | 6/27 (22.2) | 3/36 (8.3) | 1/19 (5.3) |
| Hyperprolactinemia | 21/27 (77.8)$ | 17/36 (47.2) | 7/17 (41.2) |
| **Diabetes Insipidus** | 11/28 (39.3) | 11/37 (29.3) | 18/25 (72)$ # |
| **Radiological features** | | | |
| Pituitary enlargement | 25/28 (89.3) | 31/37 (83.8) | 16/25 (64) |
| Thickened stalk | 17/22 (77.3) | 33/35 (94.3) | 23/24 (95.8) |

Data are expressed as % (n/N) or median (interquartile range).
*p value of <0.05 between < 6.5 and > 6.5-12 weeks of treatment with glucocorticoids.
#p value of <0.05 between > 6.5-12 and > 12 weeks of treatment with glucocorticoids.
$p value of <0.05 between < 6.5 and > 12 weeks of treatment with glucocorticoids.

**Supplementary table 2A: Comparison of clinical, hormonal, and radiological outcomes in severe cases according to route, dose, and duration of glucocorticoid treatment.**

| | Route | | Dose | | | Duration | | |
|---|---|---|---|---|---|---|---|---|
| | Oral (N=40) | Intravenous (N=24) | Medium dose (N=10) | High dose (N=21) | Very high dose (N=20) | <6.5 weeks (N=12) | 6.5-12 weeks (N=20) | >12 weeks (N=11) |
| **Overall response** | | | | | | | | |
| Clinical | 25/35 (71.4) | 20/22 (90.9) | 2/6 (33.3) | 20/20 (100)[b*] | 17/19 (89.5)[b$] | 9/11 (81.8) | 14/19 (73.7) | 6/9 (66.7) |
| Hormonal | 12/26 (46.2) | 12/16 (75) | 5/7 (71.4) | 8/13 (61.5) | 10/14 (71.4) | 4/7 (57.1) | 8/17 (47.1) | 6/8 (75) |
| Radiological | 17/25 (68) | 12/20 (60) | 3/5 (60) | 11/16 (68.8) | 11/19 (57.9) | 4/8 (50) | 9/15 (60) | 3/8 (37.8) |
| **Clinical response** | | | | | | | | |
| Headache | 25/29 (86.2) | 20/20 (100) | 4/6 (66.7) | 18/18 (100)[b*] | 17/17 (100) [b$] | 8/9 (88.8) | 17/19 (89.5) | 7/8 (87.5) |
| Visual defect | 12/22 (54.5) | 11/12 (91.7)[a*] | 1/5 (20) | 9/9 (100) [b*] | 10/11 (90.9)[b$] | 3/5 (60) | 7/12 (58.3) | 1/4(25) |
| Cranial nerve palsy | 3/6 (50) | 7/8 (87.5) | 0/0 (0) | 1/1 (100) | 6/7 (85.7) | 2/2 (100) | 5/8 (62.5) | 0/0 (0) |
| **Hormonal response** | | | | | | | | |
| Corticotroph | 4/15 (26.7) | 8/12 (66.7)[a*] | 0/4 (0) | 4/10 (40) | 7/11 (63.6)[b$] | 1/5 (20) | 5/9 (55.5) | 3/7 (42.9) |
| Thyrotroph | 7/16 (43.8) | 5/10 (50) | 4/5 (80) | 3/10 (30) | 5/10 (50) | 1/4 (25) | 5/8 (62.5) | 4/7 (57.1) |
| Gonadotroph | 9/16 (56.3) | 5/10 (50) | 3/3 (100) | 5/10 (50) | 5/10 (50) | 2/5 (40) | 4/6 (66.7) | 5/8 (62.5) |
| Somatotroph | 5/8 (62.5) | 2/4 (50) | 1/1 (100) | 3/3 (100) | 2/4 (50) | 2/2 (100) | 1/1 (100) | 1/4(25) |
| Prolactin normalization | 6/7 (85.7) | 5/6 (83.3) | 3/4 (75) | 3/3 (100) | 5/6 (83.3) | 1/3 (33.3) | 5/5 (100)[c*] | 2/2 (100) |
| **Diabetes Insipidus recovery** | 6/11 (54.5) | 1/5 (20) | 2/3 (66.7) | 4/7 (57.1) | 0/4 (0) | 2/3 (66.7) | 0/2 (0) | 2/5 (40) |
| **Radiological response** | | | | | | | | |
| Regression in pituitary mass | 21/24 (87.5) | 16/20 (80) | 4/5 (80) | 14/15 (93.3) | 15/19 (78.9) | 5/7 (71.4) | 12/15 (80) | 8/8 (100) |
| Resolution of stalk thickening | 12/19 (63.2) | 10/16 (62.5) | 3/5 (60) | 8/12 (66.7) | 10/16 (62.5) | 3/5 (60) | 9/14 (64.3) | 3/8 (37.8) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Need for rescue therapy** | 13/35 (37.1) | 6/23 (26.1) | 5/10 (50) | 5/17 (29.4) | 5/20 (25) | 3/12 (25) | 5/19 (26.3) | 4/8 (50) |

a*, p<0.05 between oral vs intravenous glucocorticoids; b*, p value of <0.05 between medium and high dose glucocorticoids; b#, p value of <0.05 between high and very high dose glucocorticoids; b$, p value of <0.05 between medium and very high dose glucocorticoids; c*, p value of <0.05 between and 6.5-12 weeks duration glucocorticoid treatment

**Supplementary table 2B: Comparison of clinical, hormonal, and radiological outcomes in mild-moderate cases according to route, dose, and duration of glucocorticoid treatment.**

| | Route | | Dose | | | Duration | | |
|---|---|---|---|---|---|---|---|---|
| | **Oral (N=51)** | **Intravenous (N=31)** | **Medium dose (N=6)** | **High dose (N=22)** | **Very high dose (N=28)** | **<6.5 weeks (N=15)** | **6.5-12 weeks (N=37)** | **>12 weeks (N=14)** |
| **Overall response** | | | | | | | | |
| Clinical | 22/25 (88) | 13/14 (92.9) | 5/6 (83.3) | 7/9 (77.8) | 11/12 (91.7) | 5/6 (83.3) | 21/22 (95.5) | 5/6 (83.3) |
| Hormonal | 18/39 (46.2) | 13/22 (59.1) | 2/6 (33.3) | 7/16 (43.75) | 5/19 (26.3) | 5/11 (45.4) | 17/31 (54.8) | 6/10 (60) |
| Radiological | 23/30 (76.7) | 16/27 (59.3) | 5/6 (83.3) | 16/21 (76.2) | 14/24 (58.3) | 8/13 (61.5) | 14/21 (66.7) | 12/14 (85.7) |
| **Clinical response** | | | | | | | | |
| Headache | 22/25 (88) | 12/13 (92.3) | 4/5 (80) | 7/9 (77.8) | 11/12 (91.7) | 5/6 (83.3) | 21/22 (95.5) | 5/6 (83.3) |
| **Hormonal response** | | | | | | | | |
| Corticotroph | 5/17 (29.4) | 8/16 (50) | 0/4 (0) | 4/7 (36.4) | 8/14 (57.1)[b$] | 1/8 (12.5) | 8/14 (57.1) | 3/5 (60) |
| Thyrotroph | 7/20 (35) | 5/12 (41.7) | 2/4 (50) | 7/13 (53.8) | 5/10 (50) | 0/4 (0) | 6/13 (46.2) | 4/7 (57.1) |
| Gonadotroph | 7/21 (66.7) | 7/16 (43.8) | 4/6 (66.7) | 5/14 (35.7) | 6/13 (46.2) | 3/7 (42.9) | 7/16 (43.8) | 3/8 (37.5) |
| Somatotroph | 3/7 (42.9) | 0/9 (0) | 1/1 (100) | 1/6 (16.7) | 0/7 (0) | 0/6 (0) | 2/5 (40) | 1/4 (25) |
| Prolactin normalization | 8/12 (66.7) | 8/10 (80) | 1/2 (50) | 6/8 (75) | 7/9 (77.8) | 3/7 (42.9) | 8/8 (100) | 3/4 (75) |
| **Diabetes Insipidus recovery** | 3/16 (18.8) | 7/17 (41.2) | 3/5 (60) | 2/13 (15.3) | 6/13 (46.2) | 5/6 (83.3) | 0/8 (8) | 5/12 (41.7) |
| **Radiological response** | | | | | | | | |
| Regression in pituitary mass | 15/22 (68.2) | 16/20 (80) | 5/6 (83.3) | 8/12 (66.7) | 14/18 (77.8) | 7/11 (63.6) | 15/18 (83.3) | 6/7 (85.7) |
| Resolution of stalk thickening | 19/22 (86.4) | 16/25 (64) | 4/5 (80) | 13/16 (81.3) | 14/22 (63.6) | 6/8 (75) | 15/20 (75) | 10/12 (83.3) |
| **Need for rescue therapy** | 17/44 (38.6) | 4/31 (12.9)[a*] | 5/10 (50) | 7/17 (41.2)[b#] | 3/28 (10.7)[b$] | 5/15 (33.3) | 10/36 (27.8) | 4/10 (40) |

a*, p<0.05 between oral vs intravenous glucocorticoids; b#, p value of <0.05 between high and very high dose glucocorticoids; b$, p value of <0.05 between medium and very high dose glucocorticoids; c*, p value of <0.05 between and 6.5-12 weeks duration glucocorticoid treatment

**Supplementary table 3. Predictors of anterior pituitary (AP) hormonal recovery.**

| **Parameters** | **Oral steroids** | | | **Intravenous steroids** | | |
|---|---|---|---|---|---|---|
| | **Recovery (N=30)** | **No recovery (N=36)** | **p Value** | **Recovery (N=28)** | **No recovery (N=14)** | **p Value** |
| **Baseline characteristics** | | | | | | |
| Median age | 41 (37-47) | 41 (40.5-42) | 0.974 | 38 (25.5-43.5) | 36.5 (27-48) | 0.719 |
| Female gender | 16/25 (64) | 15/21 (71.4) | 0.598 | 22/26 (84.6) | 7/13 (53.8) | 0.041* |
| Duration of symptoms (months) | 4 (2-8.8) | 8.8 (5-8.8) | 0.109 | 4 (1-8) | 14 (2-36) | 0.098 |
| **Clinical features** | | | | | | |
| Visual deficits | 4/29 (13.8) | 10/29 (34.5) | 0.065 | 6/26 (23.1) | 3/12 (25) | 0.899 |
| Cranial nerve palsy | 3/30 (10) | 5/34 (14.7) | 0.574 | 5/26 (19.2) | 1/12 (8.3) | 0.398 |
| Headache | 21/30 (70) | 22/34 (64.7) | 0.655 | 19/26 (73.1) | 6/12 (50) | 0.169 |
| **Anterior pituitary hormonal dysfunction** | | | | | | |
| Hypocortisolism | 17/25 (68) | 17/21 (80.9) | 0.326 | 23/28 (82.1) | 9/14 (64.3) | 0.207 |
| Hypothyroidism | 18/25 (72) | 18/21 (85.7) | 0.267 | 16/28 (57.1) | 7/14 (50) | 0.667 |
| Hypogonadotropism | 21/25 (84) | 18/20 (90) | 0.561 | 18/28 (64.3) | 9/14 (64.3) | 1.000 |
| Hyposomatotropism | 9/22 (40.9) | 6/20 (30) | 0.467 | 12/23 (52.2) | 9/12(75) | 0.198 |
| Hypoprolactinemia | 1/22 (4.5) | 4/18 (22.2) | 0.096 | 1/26 (3.8) | 2/12 (16.7) | 0.198 |
| ≥3 AP axes affection at baseline | 16/25 (64) | 16/21 (76.2) | 0.376 | 15/28 (53.6) | 7/14 (50) | 0.828 |
| **Diabetes Insipidus** | 12/25 (48) | 9/21 (42.9) | 0.732 | 7/28 (25) | 9/14 (64.3) | 0.015* |
| **Radiological features** | | | | | | |
| Enlarged pituitary | 22/25 (88) | 16/21 (76.2) | 0.298 | 24/28 (85.7) | 10/13 (76.9) | 0.491 |
| Thick stalk | 21/23 (91.3) | 12/15 (80) | 0.320 | 20/22 (90.9) | 12/13 (92.3) | 0.888 |

*p <0.05 is considered significant.